

Morgan Stanley Intern Interview Experience '23

About the company:

Morgan Stanley is an American multinational investment management and financial services company headquartered at 1585 Broadway in Midtown Manhattan, New York City. With offices in more than 41 countries and more than 75,000 employees, the firm's clients include corporations, governments, institutions, and individuals. Morgan Stanley ranked No. 61 in the 2021 Fortune 500 list of the largest United States corporations by total revenue.

Number of rounds : 3

- 1) Online Test
- 2) Technical Interview
- 3) HR Interview

ONLINE ASSESSMENT:

Platform: SHL AMCAT

The online assessment had 3 sections.

- 1) 10 aptitude questions to be solved in 30 minutes
- 2) 7 debugging questions to be solved in 20 minutes
- 3) 3 coding questions to be solved in 1 hour

The aptitude questions were of medium difficulty and the challenging part was to solve them within the given time. Questions were based on profit/loss, family member relations and numerical reasoning mainly. Few questions were longer than others and the trick here was to not spend too much time on a single question and move on to questions that consume less time. Your aim should be to correctly solve as many questions as possible within the given time frame. I was able to solve 8/10 questions correctly here.

As far as the debugging section was concerned we could choose between 3 programming languages- C, C++ and Java. Once we chose our language we were shown 7 code segments with syntactical, semantic and logical errors. We had to rewrite the segment of code to perform its task correctly. The important piece here is to read the question for each code segment carefully as each segment of misperforming code is expected to do something particular and achieve a

particular task. After your modifications, the code should perform the same task correctly without errors, passing all the test cases. I was able to solve 7/7 questions correctly here.

The final and all important coding section consisted of 3 questions to be solved in one hour. We could choose to code them in our preferred coding language. The questions I got were:

- 1) Given a sorted array `arr[]` consisting of N distinct integers and an integer K , the task is to find the index of K , if it's present in the array `arr[]`. Otherwise, find the index where K must be inserted to keep the array sorted.
- 2) Given an array of N numbers and a positive integer K . The problem is to find K numbers with the most occurrences, i.e., the top K numbers having the maximum frequency. If two numbers have the same frequency then the number that appears first from the beginning should be given preference. The numbers should be displayed in decreasing order of their frequencies. It is assumed that the array consists of at least K numbers.
- 3) Given an array `arr[]` of integers, find out the maximum difference between any two elements such that the larger element appears after the smaller number.

The questions were of easy-medium difficulty and I was able to pass all the test cases in all three questions.

TECHNICAL INTERVIEW

This round went on for an hour and a half and was a face to face interview conducted physically at CUIC. 14/350 students were shortlisted for the same. The interview strayed away from my conventional notion of how an interview went and was intuitive and different.

The interviewer began by asking for my introduction followed by which he straightaway jumped into a game of Tic Tac Toe following which he asked me to design the Tic Tac Toe game using OOPS. I suggested using an interface with 2 abstract methods: one to make a move and another to check if a player wins after making his/her move. He then proceeded to ask me what the minimum number of function calls would be to check if a player has won. I initially said 6 after which with his help I was able to prove that it was 4. He then modified the problem to make it a $n \times n$ board instead of the conventional 3×3 Tic Tac Toe board and then asked me to come up with a generalized expression in terms of n for the minimum number of function calls required. I was able to do this following which he modified the problem further, adding an additional rule which says that whenever a player puts a 'O' or 'X' they must also remove a 'O' or 'X' provided the board has at least 3 of their symbols at any instance. That is, at any point of time in the game there should be a maximum of 3 of my symbols on the board. He asked me what the minimum number of function calls would be in this case.

Following this the interviewer asked me some questions on OOPS concepts. I was asked what encapsulation was and also to demonstrate the need for encapsulation using an example and what would happen if encapsulation didn't exist. He then proceeded to questions on Java, asking me how I would pass command line arguments in Java as well as quizzing me on Containers in Java. He then asked me a coding question related to sorting. The question follows.

Given a train ticket consisting of multiple fields such as booking time, arrival time, departure time and ticket ID, assume the tickets are already sorted based on the basis of their ID. Ask questions to determine which sorting algorithm to use. Take the parameter based on which to sort the tickets as an input and code the sorting algorithm.

I initially suggested merge sort for which he told me I hadn't asked all the necessary questions and helped me realize that quicksort was the best option given that he wanted the code to be more space efficient than time efficient. I then proceeded to write the pseudocode for the same on pen and paper. He then asked me about hashing collision and collision resistance, which are DSA concepts based on hashing.

The interviewer then moved to Python and ML based questions since I had mentioned it in my resume. He asked me about adversarial attacks in deep learning since I was working on a research paper based on that. He also asked me basic questions on Python such as the difference between lists and tuples and about Type Error. Finally he moved to OS related questions, asking me what a batch operating system was and what were the advantages of Linux over Windows and vice versa.

After this he asked me if I had any questions and concluded the interview.

HR INTERVIEW

The HR interview went on for an hour and mostly focussed on behavioral and scenario based questions. I was asked team based situational questions such as how I would react in conflict scenarios and how I would make my teammates understand my point of view. I was also asked questions with regards to my project and if I had stuck to deadlines on all the projects. Technical questions were also asked with regards to some of my projects and I was asked how I would do things differently in my future projects. Other than this I was asked a few other basic HR questions following which the interview concluded.

Around 10:30 the same night I got an email telling me that I was selected for the internship at Morgan Stanley. 5 students were finally selected out of the 14 shortlisted for interviews.

IMPORTANT TIPS

- For preparations, stick to a few good resources instead of practicing problems on multiple platforms.
- Stick to Leetcode and solve problems from any sheet such as neetcode.io on a daily basis without fail. 5-6 problems a day starting about 3 months before the online test round is a must in order to maintain consistency. Always begin by coding the brute force approach and proceed gradually to the most optimized solution.
- Remember to revise your previously solved problems. Don't memorize the code but rather revise the approach you used and code it again from scratch.
- For OS, DBMS go through your semester notes as well as GeeksForGeeks last minute notes.
- During the interview, remember to remain calm under all circumstances! Do not get demotivated by any kind of corrections or interruptions from the interviewer's side. Don't assume that the answer you gave was incorrect or not satisfactory just because of an interruption on the interviewer's side. Be confident with the answers you give and do not get swayed.
- If you do not know an answer to a problem based question, outline your thought process and what you understand from the question. This will help the interviewer gauge your understanding of the problem statement and help you go in the right direction accordingly. Never keep silent during the interview.
- If you do not know the answer to a direct theory based question, let the interviewer know that politely rather than say something you are not sure of and make yourself prone to follow up questions.
- As far as the HR round is concerned, be completely honest and make sure to read up on the company's core values and leadership principles beforehand.
- **Last but most importantly, STICK TO YOUR PLAN.** Never ever get intimidated by what others are doing or how others are preparing or what internships others are getting. You are the product of your hard work alone and be confident of your preparation and NEVER get demotivated by a few negative results. At the end of the day there are many factors at play in deciding whether you get an internship and sometimes all of these may not be in your control. Your responsibility is just to focus on your work and do your best. Then there is no mountain that is too high to climb :)

All the best!

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